

CS5013 Project Proposal

Even Hand: A Grading-Consistency Audit for IIT Madras Courses

Mohamed Zayaan S, CE23B092 Pruthviraj Milind Chougale, CE23B104

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1 Team information

Team ID: team- God's Eye.

- Member 1: Mohamed Zayaan S – roll no. CE23B092 – Civil Engineering – GitHub handle : Zayaan3019
- Member 2: Pruthviraj Milind Chougale – roll no. CE23B104 – Civil Engineering – GitHub handle : PruthvirajChougale

Primary contact email: ce23b092@smail.iitm.ac.in.

2 Problem statement

Large courses at IIT Madras are marked by teams of teaching assistants who divide the scripts among themselves and mark independently. This introduces a component of a student's score that has nothing to do with the student: **part of what you are awarded depends on whose pile your script landed in.**

That is not speculation. The measurement literature reports it consistently: graders differ systematically in severity even from a shared rubric; a grader's severity *drifts* within a session, with marks falling as they work deeper into a stack; and markers are not self-consistent, awarding a different score to the same script weeks later. None of this is a failure of diligence – these are properties of human judgement under repetition, and every large examination board corrects for them. **No course at this institute does, because nobody has ever measured the size of the effect here.**

You cannot check it by comparing graders' averages. If A's scripts average 72 and B's average 65, two explanations fit equally well: B marks seven points harder, or B was handed the weaker half of the class. **Grader severity and student ability are confounded**, and separating them depends entirely on *how the marking was divided up* – which is where this becomes a problem about a platform rather than about one analysis.

Why this matters more at IIT Madras than elsewhere

IIT Madras grades relatively. Boundaries are computed from the cohort itself, anchored at the class average: B at the average, A one standard deviation above, S at two, then C at one below, D at two, E at three, U a fail. Alongside this, **the top 10% of the class are awarded S**. A student's grade is a function of where they sit in the distribution, not of what they scored.

Two rules govern S, and their interaction sharpens our central claim. A normal distribution places roughly 2% of a cohort beyond two standard deviations, whereas the fixed-fraction rule admits 10%.

In most cohorts the top-10% rule therefore binds, making S a rank-based award outright – the number of S grades is fixed before a single script is marked.

Three consequences follow, none of which arises under absolute grading.

1. **Severity is transferred, not merely suffered.** Under fixed cutoffs a harsh grader costs one student marks and affects nobody else. Under a relative rule the misplacement is zero-sum: a leniently marked student who rises into a band displaces a harshly marked one out of it. **For S this is an identity rather than an approximation** – the count is fixed by the 10% rule, so every S awarded to a leniently marked student is an S demonstrably taken from someone else. For the standard-deviation bands the argument is nearly as strong but not exact, since correcting the marks also moves the mean and spread that define those boundaries; we compute that shift rather than assume it away.
2. **The cutoffs are computed from a contaminated statistic.** The standard deviation placing the boundaries is meant to describe the spread of student ability. What is observed is ability spread *plus* grader-severity spread plus noise. The boundaries are set from a number that grading inconsistency has inflated, and nobody has estimated by how much.
3. **With B anchored at the mean, the busiest part of the distribution sits on a grade boundary.** Student density peaks at the average, so a correction of a mark or two moves more people across the B line than across any other. **We predict transfers will concentrate at the B/C and A/B boundaries rather than the tails** – a concrete prediction, made before seeing data, that the results can contradict.

Why this must be a platform, not a study

A tool built around one course would be useless to the next professor who asked for it, because IIT Madras courses differ in ways that change what is knowable. Class sizes run from twenty to several hundred. Some instructors mark alone; others lead teams of ten. Scores may be binary, partial-credit or ordinal. Instructors apply the relative rule, a fixed top-fraction, absolute cutoffs, or bands set at their own discretion. Most decisively, **there are two entirely different ways to divide marking, and they fail in opposite directions:**

- **Script-wise** – each assistant marks every question for a subset of students. Question difficulties remain estimable, but each grader’s severity is confounded with the average ability of the students they happened to receive. Random allocation makes that confound vanish in expectation; a fixed split by roll number does not.
- **Question-wise** – each assistant marks one question across the whole class, which is the arrangement teaching-practice guidance actively recommends and which Gradescope’s per-question assignment encourages. Every student is now seen by every grader, so abilities are cleanly linked – **but that grader’s severity appears only ever alongside that one question’s difficulty, so only their sum is identifiable.** You cannot tell a harsh grader from a hard question, and *no amount of student overlap repairs it*. The only remedy is a second marker on the same question.

The formal condition beneath both cases is connectedness: every grader and every student must be joined through some chain of shared ratings, or the model cannot place them on one scale. A professor cannot be expected to know whether their marking satisfies that. **Determining it, for whatever arrangement they happen to have used, is the first thing this platform does and the reason it generalises.**

3 Users and stakeholder

Named stakeholder: Prof. Aslam Kunhi Mohamed, Assistant Professor, Building Technology, Construction Materials and Management, Department of Civil Engineering, IIT Madras. In the discussion with the professor, we have got valuable insights from him on how we could possibly approach the entire project and make it accurate and incorporate and consider certain critical factors for the case of IITM.

What he cannot do today. At the end of each semester he sits with a mark distribution and draws lines through it. He does not know how much of that distribution's spread came from his students and how much from his assistants, though the boundaries are computed directly from it. When a student contests a grade one mark below a boundary he has nothing to answer with except his own reading of the script. He has no way to tell an assistant that they are marking harder than the others without it being an accusation rather than a measurement. And when he divides the marking again next semester, he will divide it the same way, with no idea whether that arrangement made his own consistency knowable or destroyed it.

His week, before and after. Before: cutoffs set from a distribution he cannot audit, and borderline grades defended from memory. After: cutoffs set knowing what share of the spread was his assistants rather than his students, a list of twenty scripts worth a second look, and a marking arrangement for next semester chosen deliberately.

Why he is a genuine user— The marks are his, the assistants are his, the cutoff rule is the one he applies, and he is the only person who can act on any of it. This is the sense in which the course guide asks for a stakeholder, and it is a change from our August proposal, where he would have been a well-disposed observer of a tool he had no reason to run. We put four questions to him: how the marking in his course is divided, whether any script was seen by more than one assistant, how he sets his boundaries in practice, and whether he would use an estimate of how much of a student's mark came from the grader rather than the answer. His answers are in the appendix in his own words.

One commitment we asked for before starting. Because the findings concern his own course and his own assistants, we asked that the results may be reported in full whatever they show, with graders pseudonymised and the course identified only by size and structure. Better to settle that in September than discover it in November.

The other users and their competing needs

The three groups touched by this system want incompatible things, and the design is shaped by that rather than by a single notion of ‘the user.’

- **The instructor** needs a defensible number and a short list of actions. He is the only person who holds the key mapping pseudonyms to assistants, and the only one who sees the shortlist.
- **The teaching assistants** need not to be harmed by a measurement made of them. They are pseudonymised at ingest; no output ranks named individuals; and if the instructor chooses to share an estimate with an assistant, it is that assistant's own figure with its interval and nobody else's. **Severity is a property of a marking session, not a verdict on a person**, and every output is worded that way.
- **The students** are the people the system exists for and the only group that never touches it. Their identifiers never enter the database, and nothing is ever reported at the level of a named student.

Beyond one course

Prof. Kunhi Mohamed is the first user, not the only intended one. **The intended user is any IIT Madras instructor who divides marking among assistants** – which is most instructors of most large courses here. That is why nothing about a course is hard-coded: the cutoff rule, the score type and the way the marking was divided are all inputs the platform reads from the data or is told, and Section 7 tests that claim on arrangements we did not design around rather than asserting it.

4 Prior work and why it does not suffice

1. **FACETS (Linacre, 1989) and the R packages TAM, immer, sirt** – listed first because it is the most serious overlap and we do not wish to be found to have omitted it. *Overlap*: complete, at the level of the estimator. FACETS has implemented this model commercially since 1989, calibrating examinee proficiency, rater severity and item difficulty onto one logit scale; `TAM::tam.mml.mfr` does the same free of charge. **The statistical method in Section 5 is established and available, and we claim no novelty in it.** *Gap*: these are analysis programs for measurement researchers. They consume a prepared file and emit parameter tables, assuming the user already knows their design is connected and already knows what to do with a logit. They do not audit an arbitrary course’s marking arrangement and report what it can support; they do not know what the institution does with the marks afterwards, so they cannot say that a 0.4-logit severity moved four students across the A boundary; and no instructor here is going to write a FACETS control file.
2. **Gradescope and Crowdmark** – the standard tools for grading at scale. *Overlap*: substantial and genuine; shared dynamic rubrics and comment banks reduce inconsistency by construction. *Gap*: their statistics are descriptive and per question, not per rater. More pointedly, **Gradescope’s “next ungraded” queue exists precisely to ensure no two graders see the same submission**, and its per-question assignment feature encourages the question-wise split described above. Both are sensible efficiency decisions; between them they produce exactly the two data structures in which severity cannot be estimated. The market-leading tool does not merely omit this measurement – its defaults make the data incapable of supporting it.
3. **Moodle item analysis and similar built-in reports** – facility and a crude discrimination figure per question. *Gap*: no rater facet exists anywhere in the model. These reports assume the recorded score is the truth, which is the assumption under test.
4. **Manual moderation** – current practice, where an instructor spot-checks borderline scripts. *Gap*: it inspects the scripts nearest a boundary, a different set from those most likely mismarked, and offers no way to know whose marking to inspect.
5. **Automatic grading – LLM-based marking, and the in-class quiz application proposed in this course.** *Gap*: both replace the human rather than measuring the one you have, and neither helps a course already marked. The quiz project auto-scores multiple choice, so no grader exists and no severity arises; we name it to be clear there is no collision.

Stated precisely, so the claim is not larger than it is. **The estimator is not ours and is not new. The question is.** On one side is a fifty-year-old literature with working software that assumes a well-designed study. On the other is grading software in daily use whose defaults destroy the conditions that literature requires. Running between them at this institute is a relative rule that converts an unmeasured grader effect into a letter grade taken from one student and given to another. **Our contribution is to join those three for an arbitrary course: audit what its marking arrangement can support, estimate what is estimable, propagate it through that course’s own grading rule, and hand the instructor the scripts where a re-check would change an**

outcome – then test by experiment whether the list finds anything real.

5 Tech stack

Language: **Java 21** (course policy). Build: Maven via the committed wrapper. Framework: Spring Boot 3.5 with Thymeleaf for the instructor’s views, Spring Data JPA / Hibernate, SQLite. It runs on the instructor’s own machine as a local process bound to 127.0.0.1, or in a container. **No hosted deployment and no external API**; marks and identities never leave the machine they are loaded on, which is a condition of obtaining the data at all. Input is a CSV of one row per marked response – student, question, grader, score, and where available the script’s position in that grader’s sequence – with student and grader identifiers replaced by salted hashes at ingest, before anything is written to disk.

Parsing a CSV is plumbing. Six components do the work, all implemented directly in Java rather than delegated to a statistics package:

1. **Profiling.** Detect, from the data alone, how many graders there are, which splitting arrangement was used, whether any script was marked twice, whether allocation looks random, and whether scores are binary, partial-credit or ordinal. Everything downstream is configured by this rather than assumed.
2. **The identifiability audit (the centrepiece).** **The platform’s first output is not an estimate. It is a verdict on what the data can and cannot tell you.** We construct the incidence graph joining graders, students and questions through shared ratings, and determine which parameters are separately estimable: connected components tell us which graders lie on comparable scales, and rank deficiency in the design tells us which severities are inseparable from a question’s difficulty. The report is written in the instructor’s terms – *these four assistants are comparable; these two are not, because each marked only one question; here is the smallest set of additional double-marks that would make them comparable*. **Where nothing is estimable, the platform says so and stops**, which is the behaviour that makes it trustworthy on a course it has never seen.
3. **The measurement model.** A many-facet Rasch model estimating student ability, question difficulty and grader severity simultaneously by joint maximum likelihood, with a partial-credit extension where scores are not binary. The identifiability constraint is imposed deliberately rather than discovered – the parameters are determined only up to an additive shift, so the scale is anchored explicitly, using the anchoring the audit selects, and we report what it was anchored to. Standard errors come from the observed information, because a severity estimate without an interval is unusable. The same fit yields each question’s difficulty and discrimination, flagging items that were ambiguous or mis-keyed.
4. **Drift, and whether it accumulates.** Position in a grader’s sequence enters as a covariate, testing whether marks fall as a pile is worked through. Then the question that matters across a semester: **were scripts re-allocated between assessments, or split the same way each time?** Re-randomised allocation averages a student’s luck out across quizzes and the end-semester exam; a stable split by roll number means the same student meets the same grader all semester and the effect compounds. This is an empirical question we answer from the data, and it decides whether the problem is noise or systematic disadvantage falling on identifiable students.
5. **Propagation through the course’s grading rule.** Severity in logits means nothing to an instructor. We reconstruct the mark distribution as it would have been without grader effects,

apply the course’s own rule to both, and report **the grades that moved and between whom**. The rule is pluggable – relative mean-and-standard-deviation, fixed top-fraction, absolute cutoffs, or instructor-set bands – because IIT Madras instructors use all four. Because correcting marks also moves the mean and spread, boundaries are recomputed rather than held fixed. Transfers are reported separately for the S band, where the fixed 10% share makes the exchange exact, and for the standard-deviation bands where it is not, together with the share of the class standard deviation that was grader variance rather than student ability. Each is an interval obtained by propagating parameter standard errors, never a single count.

6. **Acting on it.** Two outputs. A **re-grade shortlist** ranked not by size of correction but by **probability of changing a letter grade**, combining the correction with the student’s distance from a boundary that is itself moving – short enough that an instructor will work through it, with every entry carrying its interval and the boundary it is near. And a **linking design for next time**: the smallest set of scripts that, if double-marked, would make every grader’s severity estimable to a stated precision.

Work split and the interface between the halves. Mohamed Zayaan takes profiling, the identifiability audit, the measurement model and drift; Pruthviraj takes the grading-rule propagation, the shortlist, the linking design and the instructor’s views. The two halves meet at a single typed record – a *fitted model* carrying the three parameter vectors, their standard errors, the anchoring used and the audit’s verdict – which the first produces and the second consumes. Every component has at least one test that runs without a database, and the estimator is tested against **synthetic data with known parameters**: we generate scores from severities we chose, refit, and check the estimates recover them within their stated intervals. Both of us review the other’s code weekly from week one, since the final viva examines each of us on the other’s modules.

6 AI-usage statement

We will use AI coding assistants (Claude Code in particular) during this project, in line with the course policy on AI-assisted programming. Every line of code we submit will be code we understand and can explain in the viva, whether we typed it or the assistant generated it. No part of the estimation is delegated to a hosted model or a statistics library – the components in Section 5 are ours, and are examinable.

7 Verification plan

Recovery on synthetic data, first. Before any real mark is touched we generate score matrices from parameters we choose, refit, and report how well the true values are recovered and how that degrades as overlap shrinks. This establishes the estimator works before it says anything about a real person, and produces the curve that justifies the linking design.

Generalisation, demonstrated rather than asserted. A platform claiming to handle any course must be shown to. We construct synthetic datasets in **each** of the arrangements described in Section 2 – script-wise with random allocation, script-wise with a fixed roll-number split, question-wise, and mixed, at several class sizes and with binary and partial-credit scoring – and verify that the audit reaches the correct verdict in every one, including correctly refusing the cases where nothing is estimable. **We then run the pipeline end to end on at least one real course besides Prof. Kunhi Mohamed’s**, so that generalisation is tested against a marking arrangement we did not design around.

Two tests that let a sceptic check we are not inventing an effect, because an instructor’s first and entirely reasonable question is “*how do I know you have not found a pattern in noise?*”

- **A permutation test.** Shuffle grader labels at random, destroying any real severity structure while preserving everything else, and refit many times to obtain the distribution of severity spread under the null that graders do not differ. **If the observed spread is not clearly outside that distribution we report no detectable grader effect**, and that is the answer.
- **A negative control.** Where a course has an auto-scored component we run the identical pipeline on it with the marking agent as the grader facet. No human judgement exists there, so any severity found is spurious by construction. **Finding severity where none can exist would falsify our own method**, and we would rather discover that on a control than in the viva.

The blind re-grade – the headline experiment. We take the shortlist and an equal number of scripts the model considers unremarkable, strip both of names and marks, shuffle them, and give them to the instructor to re-grade cold. The comparison and the sample size are fixed before any result is seen: **did the flagged scripts change more than the unflagged?** *If the answer is no we report that*, and will have produced an honest negative finding on a question this institute has never asked.

Acceptance criteria the stakeholder will use to say “yes, this works”:

- The audit’s verdict on his course matches what he knows about how his assistants divided the marking, including any assistant it declares incomparable.
- The spread in severity is reported *in marks*, not model units, so he learns the size of the effect in the units he grades in.
- The number of letter grades that would change is stated with its uncertainty, together with the number of transfers – reported separately for the S band, where the fixed 10% share makes the exchange exact.
- He works through the shortlist in one sitting and judges it a reasonable use of his time.
- Every severity figure carries an interval, so an assistant who marked thirty scripts is visibly less precisely estimated than one who marked three hundred.
- At least one question flagged as ambiguous is one he independently agrees was poor, and we report the flagged questions he rejects with his reasons.

The limitation we report rather than hide. A re-grade is not ground truth. The literature motivating this project finds markers are not self-consistent over time, so the blind re-grade measures agreement between two noisy readings rather than convergence on a true score. We frame every result that way and do not claim to have found the correct mark for anybody.

8 Milestone plan

- **By the design doc (11 Sep 2026):** anonymisation protocol agreed in writing; the CSV parsing; **the audit run on the real marking arrangement**, so we know before building whether severity is estimable at all; the cutoff rule he applies recorded; interface record and module boundaries specified; the synthetic generator written *before* the estimator.
- **By the mid-demo (9 Oct 2026):** profiling and the identifiability audit working across all four synthetic arrangements; the measurement model fitting real course data with anchoring and standard errors; parameter recovery shown across overlap levels; the permutation test run; severity reported in marks; git history from both members and an honest list of what is missing.
- **By the final submission (6 Nov 2026):** drift and allocation-regime analysis complete; grade-transfer counts under the real cutoff rule; shortlist and linking design delivered; the pipeline run

end to end on a second real course; the blind re-grade run and reported whichever way it falls; a handoff pack letting him re-run it next semester; and at the demo, **Prof. Kunhi Mohamed loads his own course's marks, reads out what the audit says his data can support, how many grades moved and between whom, and opens the shortlist he intends to re-check.**

9 Risks and plan B

1. **Risk:** his course is marked by him alone, or by a single assistant. **Then there is no grader facet and no project**, and no modelling rescues it. We settle this before writing anything. **Plan B:** he introduces us to a colleague teaching a course with a full marking team and remains involved as the person who framed the question. The platform is unchanged; only whose marks form the first case study changes. *This is precisely why the system is built to accept an arbitrary course rather than one.*
2. **Risk:** the marking was split question-wise, so severity is inseparable from question difficulty. **Plan B:** this is a verdict, not a failure – the audit reports it, names the affected graders, and specifies the second-marking that would resolve it. We then either obtain that second marking for one assessment, or run the case study on a course whose arrangement supports estimation, reporting the first course's audit as a result in its own right.
3. **Risk:** the class is too small, or has too few questions, for usable precision. **Plan B:** report the intervals honestly and refuse to produce a shortlist the estimates cannot support. *A system that declines when the data is too thin is a better result than one that answers anyway.*
4. **Risk:** the permutation test comes back null – severity indistinguishable from chance. **Plan B:** a legitimate outcome, treated as one. We report the null, the effect size we can rule out, and the sample that would have been needed to detect an effect large enough to matter under the relative rule. **We will not go looking for a different course until we find a positive result**, and we say so in the report.
5. **Risk:** severity estimates could damage an assistant's standing. **Plan B:** graders pseudonymised at ingest with only the instructor holding the key; no output ranks named individuals; every estimate published with its uncertainty; and the framing throughout is calibration of a marking session rather than judgement of a person. This is written into the data-sharing note the instructor countersigns before we receive anything.
6. **Risk:** he declines the blind re-grade, which costs real time. **Plan B:** negotiate down to ten flagged and ten control scripts in advance; if he declines entirely, the permutation test, synthetic recovery and the audit still stand, and we report the missing experiment as a gap rather than substituting a weaker proxy.
7. **Risk:** the scope is too large for ten weeks and two people. **Plan B:** the cut line is declared in advance – question diagnostics go first, then drift, then the linking design. The audit, the measurement model, the grading-rule propagation and the blind re-grade are never cut, because they are the project.

10 Stakeholder acknowledgement

We had spoken to him offline and presented the entire project to him but we haven't gotten an acknowledgement mail from his end yet. So, we will be attaching the screenshot of the mail as soon as we get it.